Name: Vanshika Wadhwani
Class: D20A
Roll no: 35

Experiment No: 4

AIM:

To develop a blockchain-powered web application using Solidity programming language on Remix IDE and embedding Metamask Wallet.

Theory:

Blockchain is a distributed and tamper-resistant digital ledger used to record transactions and data. In this experiment, blockchain technology is used to implement a Blockchain-Based Rental Agreement System, where rental agreement details between a landlord and tenant are stored and managed using a Solidity smart contract. The experiment uses Solidity, Remix IDE, MetaMask, and the Ethereum Sepolia Testnet.

The smart contract stores rental agreement information such as:

- Agreement ID
- Landlord's blockchain address
- Tenant's blockchain address
- Property address
- Rental amount
- Rental duration
- Agreement active status
- Agreement acceptance status

The smart contract provides functions to create a rental agreement, retrieve agreement details, allow the tenant to accept the agreement, and allow the landlord to terminate the agreement.

A major feature demonstrated is access control using smart contracts. Only the registered tenant can accept an agreement:

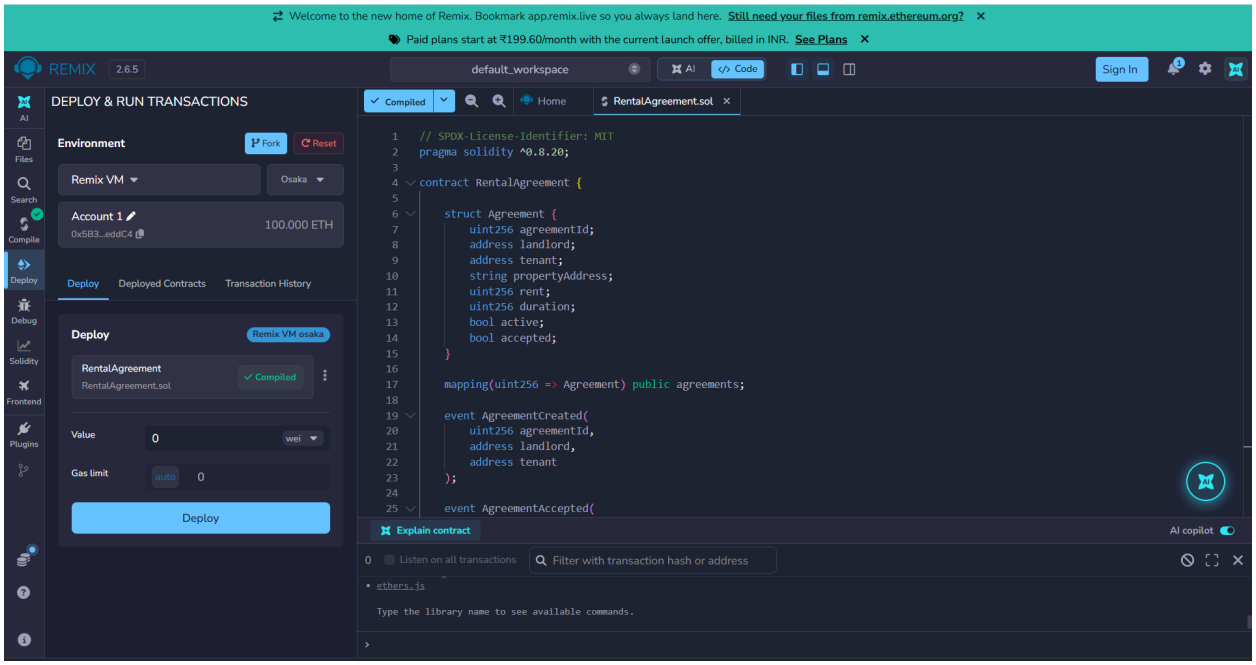
```
require(
   agreements[_agreementId].tenant == msg.sender,
   "Only tenant can accept"
);
```

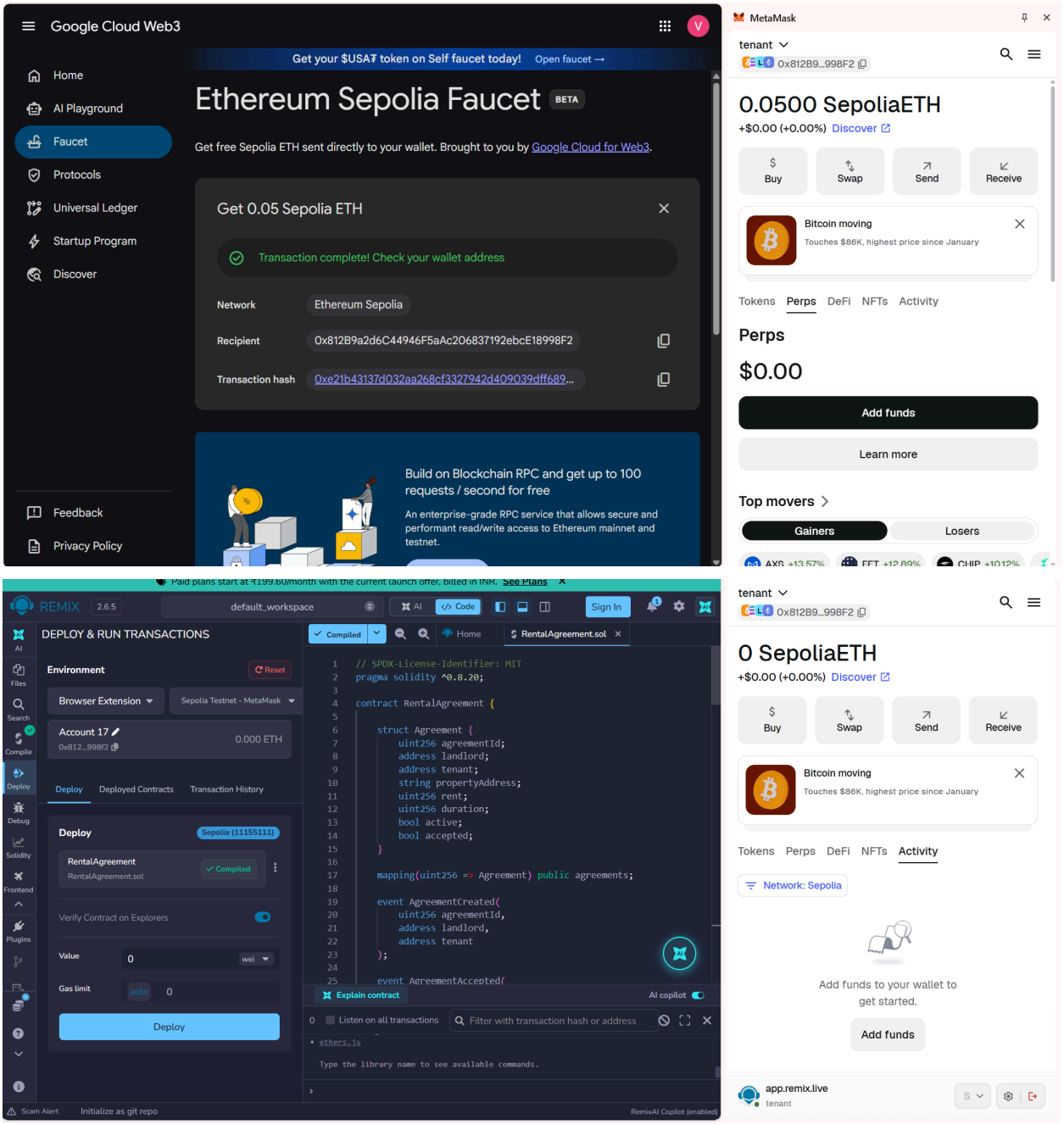
Similarly, only the landlord can terminate the agreement:

```
require(
    agreements[_agreementId].landlord == msg.sender,
    "Only landlord can terminate"
);
```

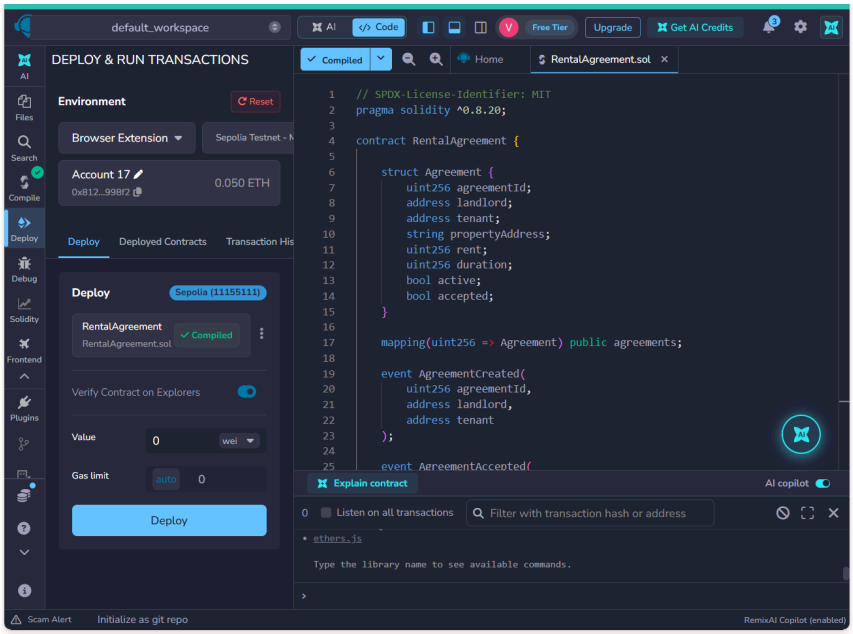
These conditions ensure that unauthorized users cannot perform restricted operations. Thus, the smart contract automatically enforces the rules of the rental agreement. This follows the same access-control concept demonstrated in the original experiment.

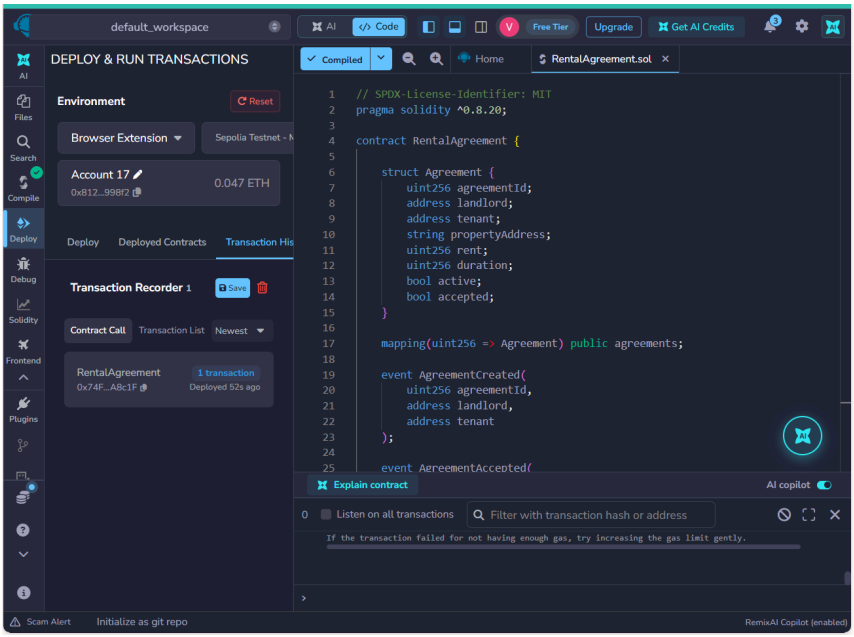
The experiment involves setting up MetaMask, connecting it to the Sepolia Testnet through Remix, creating and compiling the Solidity smart contract, deploying it, creating a rental agreement, verifying its details, accepting the agreement using the tenant account, and testing unauthorized access.

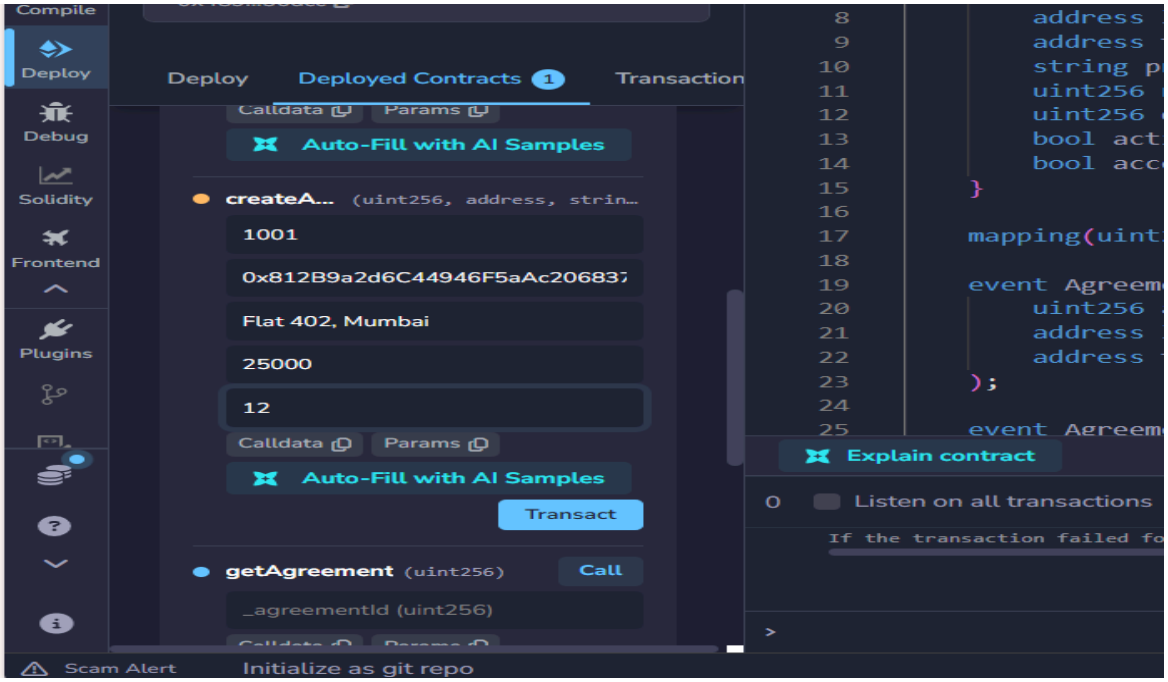



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Home
AI Playground
Faucet
Protocols
Universal Ledger
Startup Program
Discover
Feedback
Privacy Policy
Ethereum Sepolia Faucet
BETA
Get free Sepolia ETH sent directly to your wallet. Brought to you by Google Cloud for Web3.
Get 0.05 Sepolia ETH
Transaction complete! Check your wallet address
Network
Ethereum Sepolia
Recipient
0x812B9a2d6C44946F5aAc206837192ebcE18998F2
Transaction hash
0xe21b43137d032aa268cf3327942d409039dff689...
Build on Blockchain RPC and get up to 100 requests / second for free
An enterprise-grade RPC service that allows secure and performant read/write access to Ethereum mainnet and testnet.
MetaMask
tenant
0x812B9...998F2
0.0500 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Bitcoin moving
Touches $86K, highest price since January
Tokens
Perps
DeFi
NFTs
Activity
Perps
$0.00
Add funds
Learn more
Top movers
Gainers
Losers
AXS +13.57%
FET +12.89%
CHIP +10.12%

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REMIX 2.6.5
default_workspace
Sign In
DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
Sepolia Testnet - MetaMask
Account 17
0x812...998F2
0.000 ETH
Deploy
Deployed Contracts
Transaction History
Deploy
Sepolia (11155111)
RentalAgreement
RentalAgreement.sol
Compiled
Verify Contract on Explorers
Value
0
wei
Gas limit
auto
0
Deploy
Compiled
Home
RentalAgreement.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract RentalAgreement {
struct Agreement {
uint256 agreementId;
address landlord;
address tenant;
string propertyAddress;
uint256 rent;
uint256 duration;
bool active;
bool accepted;
}
mapping(uint256 => Agreement) public agreements;
event AgreementCreated(
uint256 agreementId,
address landlord,
address tenant
);
event AgreementAccepted(
Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or address
ethers.js
Type the library name to see available commands.
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)
tenant
0x812B9...998F2
0 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Bitcoin moving
Touches $86K, highest price since January
Tokens
Perps
DeFi
NFTs
Activity
Network: Sepolia
Add funds to your wallet to get started.
Add funds
app.remix.live
tenant


default_workspace
DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
Account 17
0x812...998f2
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Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or address
ethers.js
Type the library name to see available commands.
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)
tenant
Wallet 1
Deploy a contract
This site wants you to deploy a contract
Estimated changes
No changes
Network
Sepolia
Request from
app.remix.live
Network fee
0.0025 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm


default_workspace
DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
Account 17
0x812...998f2
0.047 ETH
Deploy
Deployed Contracts
Transaction His
Transaction Recorder 1
Save
Contract Call
Transaction List
Newest
RentalAgreement
0x74F...A8c1F
1 transaction
Deployed 52s ago
Compiled
Home
RentalAgreement.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract RentalAgreement {
struct Agreement {
uint256 agreementId;
address landlord;
address tenant;
string propertyAddress;
uint256 rent;
uint256 duration;
bool active;
bool accepted;
}
mapping(uint256 => Agreement) public agreements;
event AgreementCreated(
uint256 agreementId,
address landlord,
address tenant
);
event AgreementAccepted(
Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or address
If the transaction failed for not having enough gas, try increasing the gas limit gently.
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)
tenant
0x812B9...998F2
0.0473 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Bitcoin moving
Touches $86K, highest price since January
Tokens
Perps
DeFi
NFTs
Activity
Network: Sepolia
Today
Contract interaction
With Contract
app.remix.live
tenant



Sepolia Testnet

Search by Address / Txn Hash / Block / Token /

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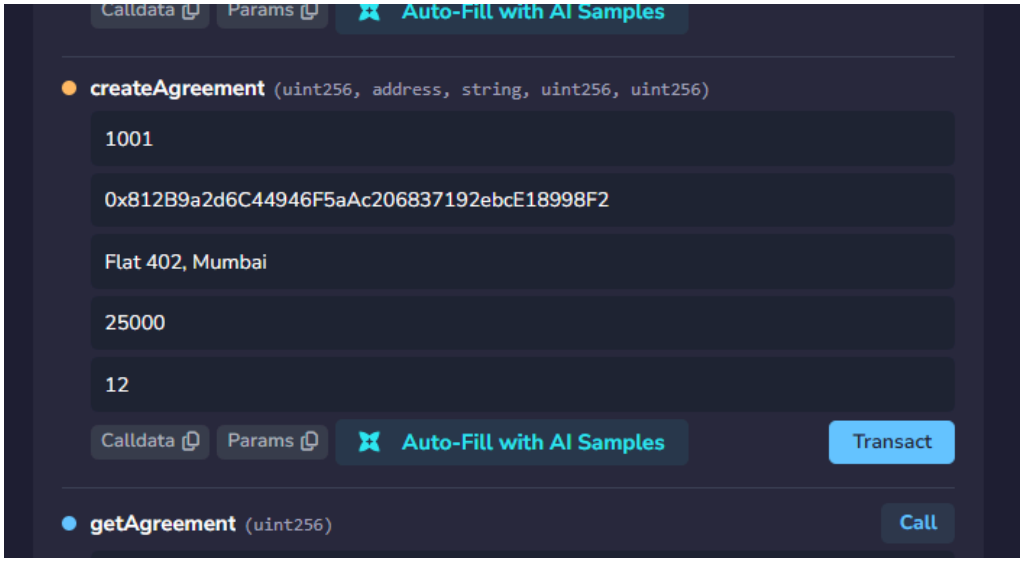
Overview | State

</> API

TRANSACTION ACTION
Contract Created at 0x1Faa2941...eC60B6C1D

[This is a Sepolia **Testnet** transaction only]

Transaction Hash:	0xad7177b41bfc7da8be37b098d6a12f5c3c823f576ef1ee8422484c7db619af0d
Status:	Success
Block:	11844116 92 Block Confirmations
Timestamp:	20 mins ago Oct-04-2026 07:33:36 PM +UTC
From:	0x812B9a2d6C44946F5aAc206837192ebcE18998F2
To:	[0x1faa294150537849b2597fc0817af93ec60b6c1d Created]
Value:	0 ETH
Transaction Fee:	0.0026615626579164 ETH
Gas Price:	2.608288368 Gwei (0.000000002608288368 ETH)

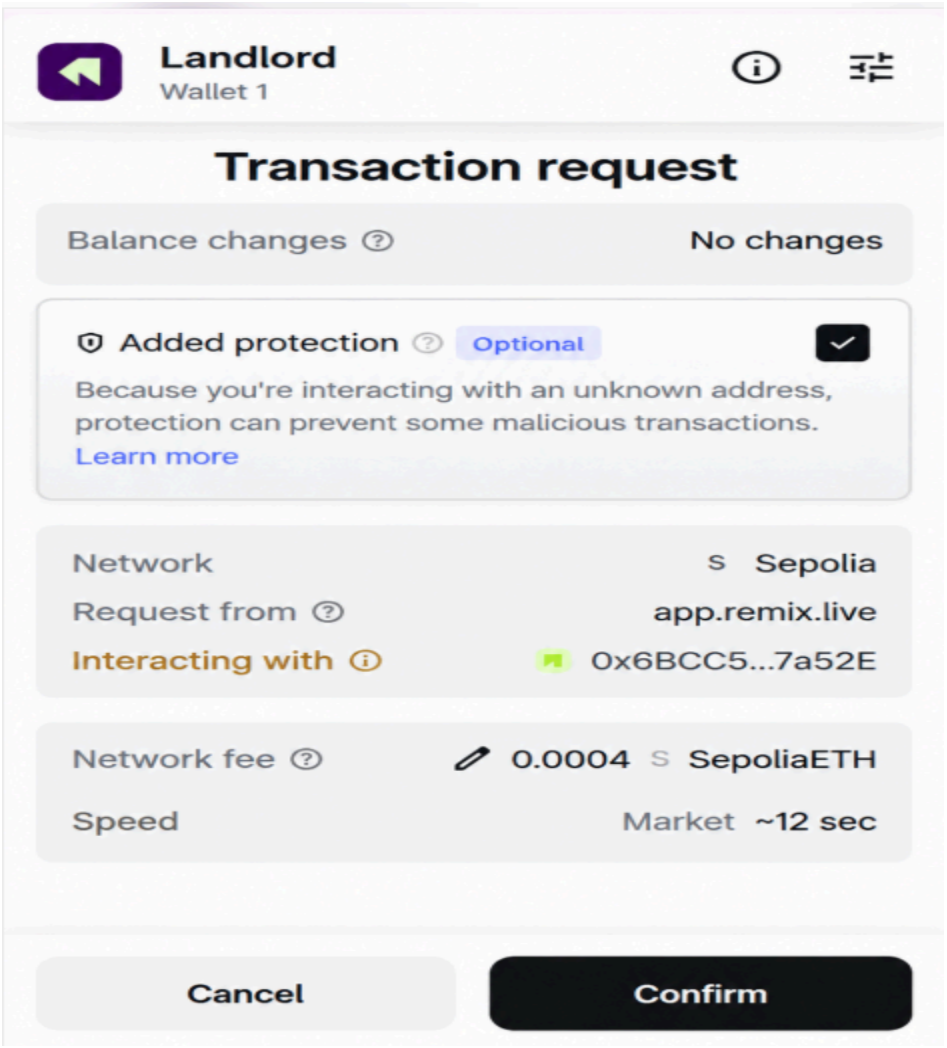


Gas estimation failed

Gas estimation errored with the following message (see below). The transaction execution will likely fail. Do you want to force sending?

missing revert data (action="estimateGas", data=null, reason=null, transaction=
{ "data":
"0x608060405234801561000f575f5ffd5b506111858061001c5f395ff3fe6080
6040523480156100f575f5ffd5b5060043610610055575f3560e01c80633a
e776d2146100595780634f9f6fe614610075578063925a9d1c146100ac5780
639a026924146100c8578063bd14de96146100e4575b5f5ffd5b610073600
480360381019061006e9190610907565b61011b565b005b61008f6004803
60381019061008a9190610907565b610282565b6040516100a398979695
9493929190610a0a565b60405180910390f35b6100c66004803603810190
6100c19190610be3565b610479565b005b6100e2600480360381019061006100
dd9190610907565b6106d6565b005b6100fe6004803603810190610f991
90610907565b6107a3565b604051610112989796959493929190610a0a56
5b60405180910390f35b3373ffffffffffffffffffffffffffffffffffffffff165f5f838152602
0019081526020015f206002015f9054906101000a900473ffffffffffffffffffffffff
ffffffffffffff1673ffffffffffffffffffffffffffffffffffffffff16146101bb576040517f08c379a
0000000000000000000000000000000000000000000000000000000081
526004016101b290610cc0565b60405180910390fd5b5f5f8281526020019
081526020015f206006015f9054906101000a900460ff1661021a57604051
7f08c379a0000000000000000000000000000000000000000000000000000000
0000081526004016102119061d28565b60405180910390fd5b60015f5f8
381526020019081526020015f2060060160016101000a81548160ff021916
9083151502179055507ff224101ad9200623928d9e682ba2635e7cc131c09
27bce7b9c0ae8a8cb21879381336040516102779291906l0d46565b60405
180910390a150565b5f5f5f60605f5f5f5f5f5f8b81526020019081526020
15f206040518061010001604052908l5f82015481526020016001820l5f90
54906101000a900473ffffffffffffffffffffffffffffffffffffffff1673ffffffffffffffffffffffffffffffff
ffffffff1673ffffffffffffffffffffffffffffffffffffffff16815260200160028201 5f90549061

Send Transaction | Cancel Transaction



CONCLUSION:

The Blockchain-Based Rental Agreement System was successfully implemented using Solidity and Remix IDE. The smart contract was deployed on the Ethereum Sepolia Testnet using MetaMask. A rental agreement was created with landlord, tenant, property, rent, and duration details. The agreement details were successfully retrieved and verified, and the tenant was able to accept the agreement using the authorized blockchain account.

The access-control mechanism was also demonstrated by restricting agreement acceptance to the registered tenant and agreement termination to the landlord. Therefore, the experiment successfully demonstrates how blockchain and smart contracts can provide transparent, tamper-resistant agreement records and automatically enforce authorization rules. This follows the original experiment's successful deployment, interaction, verification, and access-control workflow.